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APPLIED RESEARCH

A Comparative Study of Lightweight Convolutional Models for Visual Surface Material Classification

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ABSTRACT Surface material classification is essential for robotic perception, enabling grasp adaptation, slip prediction, and safe manipulation. This work presents a systematic analysis of accuracy-efficiency trade-offs among lightweight convolutional neural networks for visual-only surface material classification on the LMT-108 database. The evaluation spans both classical and compact models, including AlexNet, ResNet-18, ShuffleNetV2 (x0.5, x1.0, x1.5), MobileNetV2, MobileNetV3 (Small and Large), and EfficientNet (B0–B2), all trained under an identical pipeline and assessed in terms of accuracy, parameter count, floating-point operations (FLOPs), and inference latency. We apply a lightweight augmentation and regularization strategy to enhance generalization on texture-rich visual data. Across all architectures, EfficientNet-B2 achieved the highest mean Top-1 accuracy (97.8%). However, multi-objective analysis reveals that ShuffleNetV2 and MobileNet architectures provided the best accuracy-efficiency balance, attaining $\approx 97\%$ accuracy while reducing parameters and latency by more than 70% relative to larger models. These results highlight the suitability of modern lightweight CNNs for computationally efficient surface material classification. A supplementary haptic fusion experiment further demonstrates that top lightweight visual backbones extend naturally to compact multimodal perception systems.

INDEX TERMS Convolutional neural networks, deep learning, lightweight architectures, surface material classification.

I. INTRODUCTION

Perceiving and interpreting surface materials is a fundamental component of embodied intelligence [1], [2]. For robotic systems, the ability to recognize and characterize surfaces enables autonomous systems to infer object properties, adapt grasping strategies, and navigate unstructured environments [3], [4], [5]. Whether an autonomous entity must determine the slipperiness of a metallic component or distinguish carpet from tile, surface material classification forms the foundation for numerous downstream tasks in

robotics, industrial inspection, and assistive technologies [1], [6], [7].

Early computational approaches relied on engineered visual texture descriptors such as filter bank features [8], Local Binary Patterns (LBP) [9], and co-occurrence matrices [10]. Deep CNNs such as AlexNet, VGG, and ResNet significantly improved texture and material recognition by learning hierarchical representations [11], [12], [13]. However, their large parameter counts and computational cost hinder deployment in embedded systems [14], [15]. Recent architectural discoveries, including depthwise separable convolutions, channel shuffling, and compound scaling, have enabled lightweight models such as MobileNetV2 [16], ShuffleNetV2 [17], and EfficientNet-B0 [18] to maintain

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competitive accuracy while drastically reducing computation. However, their relative accuracy–efficiency trade-offs under a unified training and evaluation protocol for surface material classification have not been consistently characterized [19].

Rather than proposing a new architecture, this work presents a systematic evaluation of accuracy–efficiency trade-offs among representative lightweight CNNs for visual surface material classification on the LMT-108 dataset. Classical baselines (AlexNet, ResNet-18) and modern compact models (MobileNetV2/V3, ShuffleNetV2, EfficientNet-B0–B2) are trained under a unified pipeline and evaluated using accuracy, parameter count, floating-point operations (FLOPs), model size, and inference latency. While multimodal approaches incorporating visual, haptic, or acoustic cues can further improve performance, they introduce additional sensing, synchronization, and computational overhead that complicate deployment on resource-constrained platforms [20], [21], [22]. Establishing the upper bound of lightweight visual-only performance is an important prerequisite before introducing multimodal fusion.

The contributions of this research are as follows: (1) A systematic accuracy-efficiency evaluation of lightweight CNN architectures for surface material classification on the LMT-108 dataset is presented, where all models are fine-tuned under an identical training, augmentation, and regularization pipeline for controlled comparison; (2) an augmentation and regularization strategy tailored for low-sample, texture-rich visual data is developed to improve generalization without increasing computational cost; and (3) a lightweight visuo-tactile fusion case study is presented demonstrating efficient multimodal extension.

By emphasizing both performance and efficiency, this study aims to guide the selection of CNN backbones for texture-aware perception in contexts where compute and memory budgets are limited.

II. RELATED WORK

A. VISUAL TEXTURE AND MATERIAL CLASSIFICATION

CNN-based representations have largely replaced hand-crafted texture descriptors such as LBP and co-occurrence matrices, offering hierarchical feature extraction that generalizes across lighting and scale variations [19], [23], [24], [25]. Early CNNs such as AlexNet and VGG were later surpassed by residual architectures like ResNet, which improved training stability and accuracy on fine-grained material datasets. Mandal et al. [26] demonstrated that ImageNet-pretrained CNNs achieve high texture recognition accuracy even without dataset-specific architectures, though their computational demands remain significant. Despite these successes, systematic analyses of compact CNNs on material-specific datasets such as LMT-108 remain limited [20], [24].

B. HAPTIC AND MULTIMODAL SURFACE PERCEPTION

In parallel with visual approaches, multimodal perception frameworks have leveraged tactile, acoustic, and other

sensing modalities to improve surface understanding [21], [22], [24], [27], [28]. Substantial accuracy gains have been demonstrated by model architectures that utilize multimodal inputs for material classification. For example, Zheng et al. [24] introduced FusionNet, a fully convolutional fusion model that integrates two sub-networks: HapticNet for tactile data, and VisualNet for image data, achieving substantially better accuracy than single-modality performance.

Building on this, subsequent studies have explored cross-modal feature learning [29], [30], domain adaptation, and ensemble fusion. Liu et al. [20] proposed an optimized ensemble approach integrating CNN, recurrent, and acoustic models with a feature-selection strategy, achieving good accuracy on LMT-108. Other researchers extended multimodal learning with graph neural networks [31], [32], 3D tactile point clouds [33], [34], or attention-based fusion [3], [35]. Despite these advances, multimodal frameworks incur higher inference latency, sensor synchronization challenges, and hardware costs, factors that hinder deployment on small robotic platforms [21], [22], [24]. Naturally, there is a motivation to examine the upper bound of purely visual systems before adding tactile complexity.

C. LIGHTWEIGHT EDGE-EFFICIENT CNNs

Recent lightweight architectures achieve strong accuracy–efficiency trade-offs through structural optimizations such as depthwise-separable convolutions and channel shuffling. MobileNetV2 introduced inverted residuals [16], ShuffleNetV2 balanced computation and memory through channel-split design [17], and EfficientNet applied compound scaling to optimize performance under fixed compute budgets [18].

Although these networks have demonstrated competitive performance on large-scale vision benchmarks, their comparative accuracy and efficiency metrics on fine-grained, texture-centric datasets remain underexplored. Mandal et al. [26] conducted a related comparative study on rock textures using ImageNet-pretrained CNNs and reported high recognition accuracy, but their work did not evaluate computational metrics such as parameter count, FLOPs, or inference latency. In contrast, the present study extends this comparative perspective to lightweight CNN architectures, jointly evaluating predictive accuracy and computational efficiency on the LMT-108 dataset to identify optimal trade-offs for embedded robotic perception.

III. METHODOLOGY

This work conducts a unified evaluation of lightweight convolutional neural network architectures for visual surface material classification using the LMT-108 dataset [36]. The goal is to identify compact models capable of high accuracy while remaining suitable for resource-constrained platforms.

The study benchmarks AlexNet, ResNet-18, ShuffleNetV2 (x1.0, x1.5), MobileNetV2, MobileNetV3 (Small and Large), and EfficientNet-B0–B2. All models are trained under identical preprocessing, optimization, and regularization settings

to ensure controlled comparison. The evaluation incorporates both predictive performance and computational efficiency, including parameter count, FLOPs, model size, and inference latency.

A supplementary multimodal fusion experiment further demonstrates the extensibility of the strongest lightweight visual backbone toward efficient visuo-tactile perception.

A. DATASET

All experiments utilized the LMT-108 dataset [36], a comprehensive benchmark for surface material classification developed by the Technical University of Munich. The dataset contains 108 surface classes, each representing a distinct material type. Each material was recorded under six sensing modalities: visual, haptic, acoustic, friction, metal detection, and reflectance.

In this study, only the visual modality was used, consisting of RGB images captured under two illumination conditions: with flash and without flash. Each class contains 40 images in total: 20 for training (10 with flash, 10 without) and 20 for testing (10 with flash, 10 without), resulting in 4,320 RGB images across all 108 classes. The haptic modality was incorporated only in the supplementary multimodal experiment (Section IV-B), where accelerometer-based features were fused with the visual embeddings to evaluate cross-sensory generalization.

The official dataset split was used directly, with no additional resampling or repartitioning performed.

B. PREPROCESSING AND AUGMENTATION

Images were resized to 224×224 and normalized using standard ImageNet preprocessing conventions. The normalization for each pixel channel is defined as:

$$x' = \frac{x - \mu}{\sigma} \quad (1)$$

where x represents the original pixel intensity, and x' represents the normalized value. The normalization parameters were defined as channel-wise mean $\mu = [0.485, 0.456, 0.406]$ and standard deviation $\sigma = [0.229, 0.224, 0.225]$. A balanced 50:50 split was used, yielding 2160 images for the train and test sets, with uniform class representations preserved in both subsets.

To mitigate overfitting, on-the-fly augmentation (random crops, flips, rotations $\pm 20^\circ$, perspective distortion, grayscale $p = 0.1$, random erasing) were applied independently to each image during data loading, ensuring that each epoch presented the network with a stochastically varied version of the dataset. The augmentation strategy was selected via preliminary ablation experiments and fixed across all architectures to ensure fair comparison under limited-sample conditions.

C. TRAINING CONFIGURATION

All networks were trained using cross-entropy loss with label smoothing ($\varepsilon = 0.1$), a regularization technique commonly used in deep image classification [37]. The loss function is

defined as:

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c}^{LS} \log(p_{i,c}) \quad (2)$$

where N denotes the number of samples, C the number of classes, $y_{i,c}^{LS}$ the smoothed target label, and $p_{i,c}$ the predicted softmax probability for class c . Label smoothing is applied according to:

$$y_{i,c}^{LS} = (1 - \varepsilon) y_{i,c} + \frac{\varepsilon}{C} \quad (3)$$

where $y_{i,c}$ denotes the one-hot ground-truth label and $\varepsilon = 0.1$ controls the degree of smoothing. Label smoothing redistributes a small fraction of probability mass away from the ground-truth class, reducing model overconfidence and improving generalization [37].

Optimization employed the AdamW optimizer with a learning rate of 3×10^{-4} and weight decay of 1×10^{-4} . To prevent overfitting, an early-stopping mechanism with patience equaling 50 epochs halted training once validation performance ceased to improve [38]. A ReduceLROnPlateau scheduler dynamically adjusted the learning rate (factor = 0.5, patience = 12) to maintain stable convergence.

D. EVALUATION METRICS AND BASELINES

Performance was assessed using both accuracy and efficiency-based metrics.

1) CLASSIFICATION METRICS

Top-1 accuracy served as the primary evaluation metric. All reported results correspond to the mean $\pm$ standard deviation over three independent training runs, accounting for stochastic variability arising from random initialization, data augmentation, and mini-batch shuffling. The official LMT-108 dataset was used for all experiments, with separate training and testing directories, and no resampling or repartitioning was performed across runs.

In addition to overall accuracy, class-wise performance and confusion matrices were also analyzed to assess robustness across visually distinct (e.g., textiles, wood) and visually similar (e.g., glass, metal) categories. The Top-1 accuracy is defined as:

$$Accuracy = \frac{1}{N} \sum_{i=1}^N 1(\operatorname{argmax}_c p_{i,c} = y_i) \quad (4)$$

where N is the total number of test samples, $p_{i,c}$ denotes the predicted probability for class C , and y_i is the true label of the i -th sample.

2) EFFICIENCY METRICS

Computational efficiency was evaluated through parameter count, FLOPs, model size, and inference latency—each reflecting deployability in constrained environments. For a convolutional layer l with kernel size k_l , input channels $c_{in}(l)$,

and output channels $c_{out}(l)$ the total parameters were calculated as:

$$P_{total} = \sum_{l=1}^L k_l^2 c_{in,l} c_{out,l} + c_{out,l} \quad (5)$$

where L denotes the total number of convolutional layers. The total number of floating-point operations (FLOPs) was computed as:

$$FLOPs_{total} = \sum_{l=1}^L 2h_l w_l k_l^2 c_{in,l} c_{out,l} \quad (6)$$

where L denotes the total number of convolutional layers, h_l and w_l represent the spatial height and width of the output feature map at layer l , respectively, and the factor of 2 accounts for the multiply-add operations performed during convolution. Average inference latency is measured as:

$$Latency_{avg} = \frac{1}{K} \sum_{i=1}^K t_i \quad (7)$$

where t_i is the elapsed inference time (ms/image) for the i -th forward pass and K represents the total number of repeated forward passes performed under identical hardware and runtime conditions. FLOPs and parameter counts were computed using PyTorch profiling utilities, and model size (MB) was determined from the serialized state dictionary.

3) COMPARATIVE BASELINES

The study benchmarked a range of lightweight convolutional architectures spanning multiple design generations:

- AlexNet (2012) [39]: a canonical deep CNN baseline.
- ResNet-18 (2016) [40]: residual mid-scale model balancing accuracy and simplicity.
- ShuffleNetV2 (x0.5, x1.0, x1.5) (2018) [17]: channel-shuffling architectures optimized for mobile inference efficiency.
- MobileNetV2 (2018) [41]: inverted residual design offering strong accuracy-latency trade-offs for embedded systems.
- MobileNetV3 (Small and Large) (2019) [42]: architecture-searched successors incorporating squeeze-and-excitation and h-swiss activations for further efficiency.
- EfficientNet B0–B2 (2019) [18]: compound-scaled models emphasizing optimal accuracy-to-compute scaling across multiple depths and widths.

All models were trained under identical optimization, augmentation, and evaluation settings to ensure a fair comparison across architectures. Quantitative results are presented in Section IV-A.

E. COMPUTATIONAL ENVIRONMENT

All experiments were conducted in PyTorch 2.3 on an RTX 4050 GPU. Training and evaluation were performed in Jupyter Notebook and VS Code.

Inference latency values reported in this paper correspond to desktop-side measurements on this workstation. Latency was measured using single-image inference (batch size = 1) with input tensors of size $3 \times 224 \times 224$. For each model, 100 consecutive forward passes were executed under inference mode, and the average elapsed time per image was computed. Input preprocessing was not included in the timing as inputs were provided directly in tensor form. All measurements were performed using FP32 precision.

Because no embedded devices (e.g., Jetson, Raspberry Pi) were benchmarked, the latency results should be interpreted only as relative comparisons between models, not as direct indicators of real-time performance on edge hardware. Although GPU synchronization was not explicitly enforced during timing, all models were evaluated under identical conditions. Therefore, the reported latency values remain valid for relative comparison across architectures.

Unless otherwise specified, all architectures were trained under the same preprocessing, augmentation, and optimization settings, ensuring a controlled and fair comparison across models.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. PRIMARY RESULTS

The results demonstrate that modern lightweight convolutional architectures can achieve high performance in surface material classification while substantially reducing computational cost. Among the tested models, ShuffleNetV2 $\times$ 1.0 and MobileNetV3-Small attained accuracies of 97.0% and 96.3%, respectively, closely matching the 97.0–97.8% range of the more complex EfficientNet variants (B0–B2) despite using an order of magnitude fewer parameters.

Specifically, ShuffleNetV2 $\times$ 1.0 achieved an average inference latency of 4.08 ms per image with only 1.36 M parameters, representing an 82% reduction in parameter count and a 45% lower latency compared to EfficientNet-B2 (7.85 M, 7.37 ms), while maintaining accuracy within 0.8 percentage points of the best-performing model. Table 1 summarizes accuracy and efficiency metrics for all evaluated architectures. EfficientNet-B2 achieved the highest mean accuracy (97.81%) but at the cost of 7.85 M parameters and 0.69 B FLOPs. In contrast, ShuffleNetV2 $\times$ 1.0 reached a comparable 97.04% accuracy while requiring over 80% fewer parameters and nearly half the inference latency, making it the optimal baseline for subsequent experiments. Smaller MobileNetV3 and ShuffleNetV2 variants further reduced FLOPs and storage with minor accuracy loss, an important trade-off for memory or power-constrained platforms [19].

Figure 1 summarizes model parameters and computational cost (FLOPs) for the lightweight CNN architectures (AlexNet and ResNet-18 are omitted for clarity due to their significantly larger scale). AlexNet and ResNet-18 remain the heaviest models, with 57.45 M and 11.23 M parameters and 0.71 B and 1.82 B FLOPs, respectively. EfficientNet variants show greater parameter efficiency but still incur relatively high computational cost as depth increases

TABLE 1. Performance comparison of CNN architectures.

Model	Avg. Accuracy	Params (M)	FLOPs (B)	Model Size (MB)	Inference Time (ms/img)
AlexNet	93.49%	57.45	0.71	229.79	4.94
EfficientNetB0	97.01%	4.15	0.41	16.88	5.22
EfficientNetB1	97.10%	6.65	0.6	27.04	7.35
EfficientNetB2	97.81%	7.85	0.69	31.87	7.37
MobileNetV2	96.50%	2.36	0.32	9.7	3.46
MobileNetV3	96.99%	4.34	0.23	17.57	3.98
Large					
MobileNetV3	96.31%	1.63	0.06	6.65	3.55
Small					
ResNet18	96.84%	11.23	1.82	45.01	1.32
ShuffleNetV2	96.05%	0.45	0.04	1.96	4.14
x0.5					
ShuffleNetV2	97.04%	1.36	0.15	5.63	4.08
x1.0					
ShuffleNetV2	97.18%	2.59	0.31	10.57	4.1
x1.5					

(e.g., EfficientNet-B2: 7.85 M params, 0.69 B FLOPs). In contrast, the ShuffleNetV2 and MobileNet families demonstrate substantially lower complexity. ShuffleNetV2 $\times$ 0.5 and $\times$ 1.0 require only 0.45 M / 1.36 M parameters and 0.04 B / 0.15 B FLOPs, respectively, over an order of magnitude lighter than AlexNet and ResNet-18. A generally positive correlation between model size (parameters) and computational complexity (FLOPs) can be observed, with architectural design influencing the scaling efficiency.

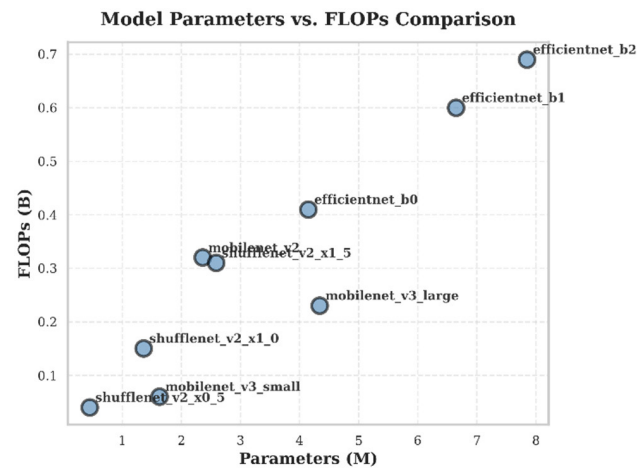


FIGURE 1. Comparison of model parameters (millions) and computational complexity(FLOPs, billions) for mobilenet, ShuffleNet, and efficientnet variants, highlighting the scaling of computational cost with network size.

Figure 2 reports the corresponding accuracy and inference latency measurements for the lightweight architectures (AlexNet and ResNet-18 are omitted for visual clarity). EfficientNet-B2 attains the highest classification accuracy at 97.81%, but this improvement is accompanied by increased latency (7.37 ms). ShuffleNetV2 has a latency of 4.08 ms, indicating a more favorable balance for real-time inference. The MobileNet models provide the lowest latency overall (3.46–3.98 ms) but show slightly reduced accuracy

relative to the ShuffleNetV2 and EfficientNet models. Although ResNet-18 incurs a large FLOP count, it exhibits fast inference (1.32 ms), likely due to efficient kernel implementations and the regular structure of its convolutional blocks. These observations show that gains in accuracy do not necessarily correspond to improved runtime behavior.

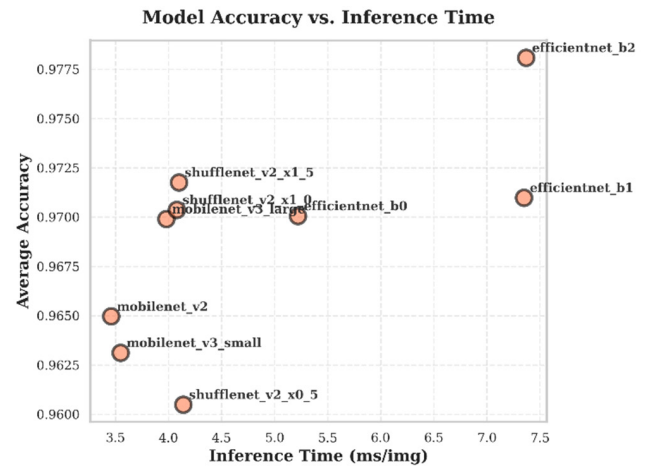


FIGURE 2. Accuracy classification accuracy versus inference time (ms/image) for MobileNet, ShuffleNet, and efficientnet architectures, illustrating the trade-off between predictive performance and computational efficiency.

Finally, Figure 3 presents a combined accuracy-efficiency map summarizing the overall trade-offs across the best-performing subset of evaluated models. Resource efficiency is defined as the ratio of average classification accuracy to inference time (ms/image), representing the accuracy achieved per unit latency. The visualization shows that ShuffleNetV2 $\times$ 1.0 occupies the upper-right region of the trade-off space, achieving both high predictive accuracy and strong computational efficiency. Other competitive architectures include MobileNetV3 and the remaining ShuffleNetV2 models evaluated. Overall, the results suggest that ShuffleNetV2 provides the most balanced configuration for texture classification on LMT-108, combining competitive performance with real-time inference capability.

To evaluate the impact of data augmentation on model generalization, a mini ablation study was conducted using ShuffleNetV2 $\times$ 1.0 as the reference architecture. Training the model without augmentations yielded a mean accuracy of 87.13%, while the augmented variant achieved 97.04% on the LMT-108 dataset. This substantial performance gap indicates that, in small-sample texture classification settings, augmentation is critical for mitigating overfitting and enabling the learning of invariant surface representations. While part of this gain can be attributed to the limited number of training samples per class, the result highlights the strong dependence of texture-based recognition performance on effective data diversity. Accordingly, augmentation is treated as a fixed component of the training pipeline in all subsequent experiments to ensure that architectural comparisons reflect

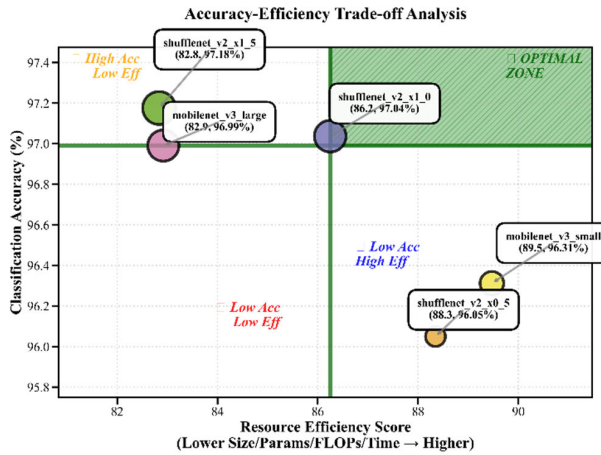


FIGURE 3. Accuracy versus resource efficiency trade-off. The optimal zone (green hatched, upper-right) identifies models achieving both high accuracy and efficiency.

representational efficiency rather than sensitivity to data scarcity.

Compared to conventional deep CNNs such as AlexNet (93.5%) and ResNet-18 (96.8%), these lightweight designs deliver 1–3 percentage points higher accuracy, while requiring substantially fewer parameters and being better suited for resource-constrained deployment scenarios. This suggests that architectural strategies such as depthwise-separable convolutions and channel shuffling can preserve representational capacity even at substantially reduced parameter and FLOP budget.

Viewed from an accuracy–efficiency perspective, ShuffleNetV2 $\times 1.0$ occupies a Pareto-efficient operating point on LMT-108 among the evaluated architectures, as no other model simultaneously achieves higher accuracy with lower parameter count or inference latency. In contrast, larger models such as AlexNet and ResNet-18 are Pareto-dominated under the evaluated metrics, offering neither improved accuracy nor reduced computational cost.

To further analyze model behavior, the most frequent misclassification pairs for EfficientNet-B2 (the highest-accuracy model) are examined in Table 2. The majority of errors occur between visually similar materials within the same material group, such as foam variants, wood species, and reflective metals. The most frequent pair, G3StainlessSteel misclassified as G3Copper, accounts for 6 out of 20 test samples for that class. A subset of errors also spans material groups (e.g., G4LaminatedWood predicted as G3Glass, G4Beech_0 as G2RoofTile), suggesting that surface reflectance and lighting conditions may contribute to confusion beyond categorical similarity. Overall, the remaining classification errors are largely attributable to inter-class visual similarity rather than systematic model failure.

B. ARCHITECTURAL DISCUSSION

The results in Table 1 and Figures 1–3 highlight how architectural design choices strongly influence the accuracy–efficiency trade-off in surface material

TABLE 2. Top confused material pairs for efficientnet-b2.

True Class	Predicted Class	Count
G3StainlessSteel	G3Copper	6
G7ProfiledFoam	G7FineFoamVersion1	3
G4LaminatedWood	G3Glass	3
G4Beech_0	G2RoofTile	3
G7StyrofoamVersion1	G7FoamFoilVersion1	2
G7EpdmFoam	G7ProfiledFoam	2
G4Teak	G4CompressedWoodVersion1	2
G4Beech_0	G4CompressedWoodVersion1	2
G4Beech_0	G4CherryTree	2
G3StainlessSteel	G4WoodenPaper	2

classification. Unlike object-centric recognition tasks that benefit from deep hierarchical abstraction, texture recognition primarily relies on the discrimination of localized, repetitive patterns and subtle differences across spatial neighborhoods [43]. Architectures that efficiently preserve and mix local features therefore exhibit a structural advantage on the LMT-108 dataset.

Classical CNNs such as AlexNet and ResNet-18 rely on dense convolutional blocks and progressive depth to increase representational power. While effective for general-purpose vision tasks, this design can lead to redundant parameterization for texture-centric problems, where increasing network depth does not necessarily increase model discriminative capacity [44]. Consequently, these architectures incur substantial computational overhead without proportional gains in classification accuracy.

EfficientNet demonstrates strong representational capacity through compound scaling of depth, width, and resolution [18]. However, this compound scaling strategy implicitly favors deeper feature hierarchies, which results in diminishing returns for small-scale texture datasets. The small performance gains observed with increasing EfficientNet depth also suggest that beyond a certain capacity, adding additional layers mostly amplifies redundancy rather than improving texture classification accuracy, while increasing inference latency and memory access cost.

In contrast, ShuffleNetV2 is explicitly optimized for efficient feature reuse and low memory access overhead. ShuffleNetV2 employs channel splitting and shuffling to reduce memory access cost while maintaining cross-channel information interaction [17], resulting in strong texture classification performance despite its fewer parameters. This design enables ShuffleNetV2 $\times 1.0$ to achieve accuracy comparable to EfficientNet-B2 while requiring over 80% fewer parameters and significantly lower inference latency, indicating that effective channel mixing can compensate for reduced model depth in texture classification settings.

MobileNet architectures further emphasize efficiency through depthwise-separable convolutions and inverted residual blocks, which separate spatial filtering from channel-wise feature mixing [41], [42]. MobileNetV3 also incorporates lightweight attention mechanisms and optimized nonlinearities, resulting in low latency across all evaluated variants.

However, a notable reduction in accuracy relative to ShuffleNetV2 suggests that aggressive parameter compression may slightly limit classification accuracy when class boundaries are visually subtle.

Interestingly, ResNet-18 exhibits a relatively fast inference time despite its high FLOP count, likely due to its regular convolutional structure and optimized kernel implementations on modern hardware. This reveals that FLOPs alone are not a determining factor for a model's runtime performance, as memory access patterns and operator regularity can also influence inference latency [17]. Nevertheless, its parameter inefficiency and lack of mobile-oriented design prevents it from achieving the same accuracy-efficiency balance as newer, lightweight models.

Overall, these results indicate that architectural strategies explicitly designed for efficient feature reuse, such as channel shuffling, inverted residuals, and depthwise-separable convolutions, are particularly well suited for surface texture classification under constrained computational budgets. Among the evaluated architectures, ShuffleNetV2 $\times 1.0$ achieves a favorable balance between accuracy, efficiency, and real-time performance on LMT-108 texture scans, supporting its potential use in real-time and embedded applications.

C. SUPPLEMENTARY MULTIMODAL EXPERIMENT

To illustrate the adaptability of the optimized visual encoder, a supplementary experiment integrated ShuffleNetV2 $\times 1.0$ with a lightweight 2-D CNN-based accelerometer classifier via feature-level fusion. The resulting MultiModalNet concatenated a 1024-dimensional visual embedding and a 256-dimensional haptic embedding to form a joint representation, which was processed by a compact fully connected fusion head. To preserve computational efficiency while enabling limited domain adaptation, only the final convolutional layers of the visual backbone and the top classifier blocks of the haptic encoder were unfrozen and fine-tuned. The overall fusion mapping can be expressed as:

$$y = f_{\text{fusion}}(z) = \text{Softmax}(W_3\sigma(W_2\sigma(W_1z + b_1) + b_2)b_3) \quad (8)$$

where W_i , b_i denote the trainable weights and biases of the fusion head's linear layers and $\sigma(\cdot)$ represents the ReLU activation.

Empirical evaluation on the LMT-108 dataset yielded a 99.35% test accuracy, outperforming the unimodal ShuffleNetV2 $\times 1.0$ baseline (97.04%). While not intended as an exhaustive optimization, this result demonstrates that lightweight visual CNN backbones can naturally extend to efficient visuo-tactile perception with minimal additional computational overhead.

V. CONCLUSION

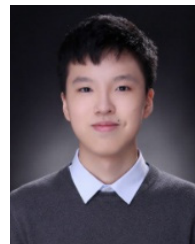
This study presented a systematic evaluation of lightweight convolutional architectures for surface material classification and showed that compact models can achieve strong performance while substantially reducing computational cost.

Although EfficientNet-B2 achieved the highest accuracy, the analysis demonstrated that ShuffleNetV2 and MobileNet variants provide the most favorable accuracy-efficiency trade-off, with ShuffleNetV2 $\times 1.0$ offering competitive performance at a fraction of the parameters and latency. The scenario-based evaluation further highlighted that no single architecture is universally optimal, emphasizing that backbone selection must align with specific deployment constraints rather than accuracy alone. These findings reduce the effective architecture search space to a small set of consistently strong candidates for each deployment scenario, enabling practitioners to select suitable backbones without exhaustive experimentation. The supplementary visuo-tactile experiment also showed that the strongest lightweight visual backbone generalizes effectively to multimodal fusion with minimal additional computational overhead. Future work will explore improved augmentation strategies, dynamic feature integration mechanisms for lightweight architectures, and hardware-level benchmarking on embedded platforms.

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